Back **Assessment 3**
Graded Assignment • 15 min

Due Aug 8, 11:59 PM IST

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 50%. We keep your highest score.

Next item

1. What are the benefits of using serialization in machine learning? 1 / 1 point

- [x] It ensures that a model can be saved and loaded consistently.

  Correct
  Correct! Serialization ensures consistent saving and loading of models, which is crucial for reproducibility.

- [ ] It reduces the complexity of the model architecture.
- [ ] It helps in managing large datasets efficiently.
- [ ] It improves the accuracy of models.
- [x] It allows models to be easily shared and deployed.

  Correct
  Correct! Serialization facilitates easy sharing and deployment of models in different environments.

2. What does a higher pixel count in an image generally indicate? 1 / 1 point

- ○ Less color accuracy
- ◉ Higher resolution and more detail
- ○ Faster image processing
- ○ Lower file size

  Correct
  Correct! A higher pixel count generally indicates higher resolution and more detail in the image.

3. Which Keras function converts a PIL image to a NumPy array? 1 / 1 point

- ◉ image_to_array
- ○ convert_image
- ○ array_to_image
- ○ image_convert

  Correct
  Correct! The image_to_array function in Keras is used to convert a PIL image to a NumPy array.

4. What is the primary use of the Python Imaging Library (PIL) in digital image processing? 1 / 1 point

- ◉ To manipulate and format images
- ○ To perform numerical computations
- ○ To create 3D graphics
- ○ To develop web applications

  Correct
  Correct! PIL is widely used for manipulating and formatting images in Python.

5. What is the purpose of the array_to_image function in Keras? 1 / 1 point

- ○ Resize images
- ◉ Convert NumPy arrays back to PIL-based images
- ○ Enhance image contrast
- ○ Convert PIL images to grayscale

  Correct
  Correct! The array_to_image function is used to convert NumPy arrays back to PIL-based images.